

Topic 4 – Plant Structure and Function, Biodiversity and Conservation

Candidates will be assessed on their ability to:

4.1	(i) know the structure and ultrastructure of plant cells including cell wall, chloroplast, amyloplast, vacuole, tonoplast, plasmodesmata, pits and middle lamella and be able to compare it with animal cells (ii) understand the function of the structures listed in (i)
4.2	be able to recognise the organelles in 4.1 from electron microscope (EM) images
4.3	understand the structure and function of the polysaccharides starch and cellulose, including the role of hydrogen bonds between the β -glucose molecules in the formation of cellulose microfibrils
4.4	understand how the arrangement of cellulose microfibrils and secondary thickening in plant cell walls contributes to the physical properties of xylem vessels and sclerenchyma fibres in plant fibres that can be exploited by humans
4.5	know the similarities and differences between the structures of, the position in the stem, and the function of sclerenchyma fibres (support), xylem vessels (support and transport of water and mineral ions) and phloem (translocation of organic solutes)
4.6	CORE PRACTICAL 7 Use a light microscope to: (i) make observations, draw and label plan diagrams of transverse sections of roots, stems and leaves (ii) make observations, draw and label cells of plant tissues (iii) identify sclerenchyma fibres, phloem, sieve tubes and xylem vessels and their location.
4.7	understand how the uses of plant fibres and starch may contribute to sustainability, including plant-based products to replace oil-based plastics
4.8	understand the importance of water and inorganic ions (nitrate, calcium ions and magnesium ions) to plants
RECOMMENDED ADDITIONAL PRACTICAL Investigate plant mineral deficiencies.	
4.9	CORE PRACTICAL 8 Determine the tensile strength of plant fibres.
4.10	understand the conditions required for bacterial growth
4.11	know that substances derived from plants can have antimicrobial and other therapeutic properties
4.12	CORE PRACTICAL 9 Investigate the antimicrobial properties of plants, including aseptic techniques for the safe handling of bacteria.

4.14	(i) understand that classification is a means of organising the variety of life based on relationships between organisms using differences and similarities in phenotypes and in genotypes, and is built around the species concept (ii) understand the process and importance of critical evaluation of new data by the scientific community leading to new taxonomic groupings, based on molecular evidence, including the three-domain system (Archaea, Bacteria and Eukarya)
4.15	know that, over time, the variety of life has become extensive but is now being threatened by human activity
4.16	understand what is meant by the terms <i>biodiversity</i> and <i>endemism</i>
4.17	know how biodiversity can be measured within a habitat using species richness, and within a species using genetic diversity by calculating the heterozygosity index: $\text{heterozygosity index} = \frac{\text{number of heterozygotes}}{\text{number of individuals in the population}}$
4.18	understand how biodiversity can be compared in different habitats using the formula to calculate an index of diversity (D): $D = \frac{N(N-1)}{\sum n(n-1)}$
4.19	understand the concept of niche and be able to discuss examples of adaptations of organisms to their environment (behavioural, anatomical and physiological)
4.20	(i) understand how the Hardy-Weinberg equation can be used to see whether a change in allele frequency is occurring in a population over time (ii) understand that changes in allele frequency can come about as a result of mutation and natural selection (iii) understand that reproductive isolation can lead to accumulation of different genetic information in populations, potentially leading to the formation of new species
4.21	be able to evaluate the methods used by zoos and seed banks in the conservation of endangered species and their genetic diversity, including scientific research, captive breeding programmes, reintroduction programmes and education